Bi-CovaSC: A Few-Shot Learning Model with Multi-Scale Feature and Covariance Bi-Metric Learning for Bearing Fault Diagnosis

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***Abstract*—The growth of modern industry comes with the inevitable application of electrical machines and their indispensable ball-bearing parts, which significantly improve the machinery's efficiency, stability and lifespan. To ensure the quality of the machines, one has to take account of the condition of the ball bearings, particularly to identify any possible faults, which is challenging. With that problem in hand, a deep-learning few-shot model named "Bi-CovaSC" is proposed, which could efficiently diagnose abnormalities in the ball bearings without much expertise knowledge and data provided. In this model, by utilizing the "MAFE" mechanism, the model could extract the feature patterns of different ranges and then learn their long-range relationship. To further highlight the extracted information both spatial-wise and channel-wise, the BAM attention block is employed before the information is passed through the Covariance-Bi-metric classification module. The "Bi-CovaSC" model is evaluated on the Paderborn University (PU) dataset and yielded high accuracy, which puts the model in a potential place for real-life application. The code will be available at: https://github.com/viet2715/fewshot-code**

Keywords—Few-shot learning, fault bearing diagnosis, Covariance Matrix, Mamba, Attention Block

I. Introduction

In electric motors, bearings help reduce friction between contact surfaces, thereby improving work efficiency and extending lifespan of machines. In fact, machinery often operates under varying conditions, temperatures, and loads, which can lead to failures in bearings. Such failures might increase friction, reduce efficiency, and result in risks of severe mechanical damage and economic losses. Accordingly, early bearing fault diagnosis is crucial and remains an important research topic.

Traditional approaches for bearing fault diagnosis often use pre-processing techniques such as Fourier Transform, Fast Fourier Transform, and Wavelet Transform in combination with conventional machine learning algorithms like Support Vector Machine (SVM) and K-Nearest Neighbors (KNN). For instance, Zhang et al. [1] proposed integrating SVM, permutation entropy, and Ensemble Empirical Mode Decomposition (EEMD) for bearing fault classification. Tian et al. [2] introduced a bearing faults classification method by extracting features using spectral kurtosis (SK) and cross-correlation, followed by classification using KNN and Principal Component Analysis (PCA).

Although traditional methods demonstrate relatively good performance in specific cases, they still heavily rely on expert knowledge and require specific algorithms tailored to each problem. To address this, deep learning-based approaches have been widely developed for bearing faults diagnosis, with Convolutional Neural Networks (CNNs) serving as the backbone for many image and signal processing models. For instance, Turker et al. [3] employed 1D convolutional layers at the input stage to accurately extract information from raw data without requiring expert intervention. Similarly, Zhao et al. [4] utilized multiple Wavelet Transforms to capture diverse frequency-domain features and integrated Deep Residual Networks—combining ResNet and Convolutional layers—to facilitate deeper architectures while mitigating gradient descent issues. Jiang et al. [5] introduced a multi-scale convolutional feature extraction approach, using kernels of varying sizes to enhance the model's capacity to learn diverse information. Beyond CNNs, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks excel at processing temporal data. Zhao et al. [6] proposed the LFGRU network, a hybrid model that integrates handcrafted features with deep learning-based gated recurrent unit (GRU) architectures for improved fault diagnosis. Additionally, transformer-based approaches [7] have shown significant success in bearing fault diagnosis [8] [9].

Traditional deep learning models require large amounts of data to achieve high performance. However, in bearing fault classification, collecting fault data is often highly challenging due to its limited occurrence and the high economic costs involved. Additionally, artificially

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simulated data often lacks accuracy, which consequently restricts the application of traditional deep learning models.

Traditional deep learning models often require a large amount of data to learn generalized features, thereby improving the model's performance. However, in bearing fault classification, collecting large-scale datasets is often challenging, as fault data is typically much scarcer than "healthy" data. Several methods have been proposed to address this issue.

One approach involves Generative Adversarial Networks (GANs) [10], which generate synthetic data to improve model training. However, GANs often demand substantial computational resources and rely on high-quality initial datasets, making their practical implementation difficult. Another strategy is transfer learning, where a model is pretrained on a large, related dataset before being fine-tuned with a smaller target dataset. While effective, finding an ideal pre-training dataset remains a key challenge. Another commonly used approach is transfer learning, which enables models to learn from a large dataset beforehand and then fine-tune on the target dataset. This method is highly effective when dealing with limited data. For instance, Zhu et al. [11] employed L1 regularization to prevent overfitting, followed by fine-tuning using an LSTM model. Similarly, Li et al. [12] leveraged a Deep Belief Network (DBN) to reduce data dimensionality, subsequently integrating Conv1D layers to extract essential features.

As an alternative approach, few-shot learning has shown promising performance for bearing fault diagnosis. For instance, Matching Networks [13] and Relation Networks [14] leverage correlation-based feature extraction and classification mechanisms. QSFormer [15] introduces a twobranch architecture, where one branch extracts global features while the other captures local features. These features are then processed through a Transformer block to compute correlations. Despite the effectiveness of these methods, they still primarily rely on similarity metrics such as cosine and Euclidean distance, which may struggle to capture complex relationships in the feature space. To address this limitation, Li et al. [16] proposed CovaMNet, which utilizes covariance matrices to enhance feature representation. Covariance matrices can effectively model dependencies between features, improving the generalization capability of few-shot learning models. The use of covariance-based representations has also been extended to other few-shot learning approaches, including Ensemble Net [17], Multilevel Few-shot [18], and SACovaMNet [19].

More recently, Mamba architecture (S6) [20] proposed by Gu and Dao brought significant improvements with its Selection Mechanism and Hardware-aware Algorithm. These enhancements effectively reduce parameter overhead and computational complexity while maintaining strong performance. Mamba has shown great potential as a backbone for various applications, including VMamba [21], Vision Mamba [22] in Computer Vision in general and in bearing faults analysis [23] [24].

In the current study, inspired by the success of the Mamba architecture as well as the advantages of CovMNet and MultiLevel few-shot learning, we present a simple yet effective approach for few-shot learning in bearing fault diagnosis. The key contributions of this work are as follows:

- Proposed model Correlation Channel and Spatial: The model is designed to learn effectively with a limited amount of data. It employs an end-to-end training approach in a single step, reducing computational complexity and simplifying the overall model structure.
- Proposed MAFE: This block enhances feature learning by leveraging Mamba and Axial depthwise convolution integration with multiple kernel sizes to capture both global and spatial information within each branch, improving performance even in data-scarce scenarios.
- Dual Branches for Channel and Spatial Correlation: The model incorporates two parallel branches—one for channel correlation and one for spatial correlation—combined with a shared-parameter Attention Block. This enables the model to effectively focus on relevant spatial and channel features, improving the relationship learning between the query and support set.

II. Related Work

Mamba Architecture. Mamba [20] introduced by Gu and Dao in late 2023, has demonstrated impressive performance in large-scale Natural Language Processing (NLP). The architecture has shown excellent performance when applied for computer vision tasks. One of Mamba-based methods in computer vision is VMamba [21]. In VMamba, the SS2D block is ultilized, which scans in four directions to capture generalized features with reduced computational cost and fewer parameters. This architecture is emerging as a promising backbone for many vision-based models leveraging Mamba in image processing tasks.

Covariance Matrix. Covariance Matrix was introduced in CovaMnet by Li et al. [16] for Few-shot learning. Unlike conventional approaches that rely on distance metrics such as Euclidean distance or cosine similarity, the use of the covariance matrix allows the model to capture second-order statistical relationships within the data. This enhances the ability to extract latent information for each class, thereby improving the representation of correlations between query and support sets.

Multi-level Few-shot Model. Unlike most of the traditional Few-shot models that produce the final scores based on only one single metric, the Multi-Level model proposed by Vu et al. [18] incorporated both spatial level and channel level classification modules. By having multiple classification heads like that, the model could have more material for the scoring step as well as leverage their ability to produce more generalized predictions where one score of a level could assist or adjust the other.

III. Methodology

In this paper, we proposed a few-shot model to handle the bearing fault diagnosis task. The Support set is denoted as $\mathbf{S} = \{\mathbf{S^1}, \mathbf{S^2},...,\mathbf{S^C}\}$ corresponding with the support label set $\mathbf{Y^S} = \{\mathbf{Y^{S1}},\mathbf{Y^{S2}},...,\mathbf{Y^{SC}}\}$, where C is the number of classes. Each $\mathbf{S^i}$ is presented as $\mathbf{S^i} \in R^{K\times C_i\times H_i\times W_i}$ with K be the number of samples per class (in this paper, we use $K = 1$ and $K = 5$). The Query set is denoted as $\mathbf{Q} = \{\mathbf{Q^1}, \mathbf{Q^2},...,\mathbf{Q^{n_q}}\}$, where q is the number of samples drawn from each class and $n_q = K \times q$.

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Similarly, the labels of the Query set are $\mathbf{Y^Q} = \{\mathbf{Y^{Q1}}, \mathbf{Y^{Q2}}, ..., \mathbf{Y^{nq}}\}$.

After being transformed into a 2D space, the image will utilize basic convolution layers to extract fundamental features and reduce the image's spatial dimensions. Subsequently, it will pass through the MAFE block, which combines Mamba and CNN networks to extract both local and global feature information. The BAM attention block is utilized to enable the model to learn important features, thereby enhancing the effectiveness of classification using the covariance matrix. Classification is performed using Covariance Matrix Layers, which capture the relationships between the Support set and Query set through two branches: Channel correlation learns the relationships between feature channels of the query and support samples, and Spatial correlation learns the relationships between spatial regions of the query and support features. Finally, Conv1D will be applied to generate the final scores for both spatial and channel dimensions.

The model aims to determine the correlation between the query samples and the classes in the Support set. In this paper, we propose the Bi-CovaSC model, as shown in Fig.1. First, The Short-Time Fourier Transform (STFT) can help capture important features by transforming signals from the time domain to the frequency domain, allowing representation of information in terms of time, frequency, and amplitude. It also converts 1D signals into 2D representations. The feature extraction is performed on the 2D input data using convolution layers and MAFE module, followed by the BAM Block to put attention mask on the query and support data. Classification is performed using Covariance Matrix Layers, which capture the relationships between the Support set and Query set through two branches: Channel correlation learns the relationships between feature channels of the query and support samples, and Spatial correlation learns the relationships between spatial regions of the query and support features.

A. The proposed Mamba Aggregrate Feature Extractor(MAFE) Block

The MAFE module, as shown in Fig. 2, is proposed for extracting features from 2D input data. First, the model applies four basic convolution layers to extract fundamental features while reducing the input size, thereby decreasing computational complexity. The input size is $1 \times 64 \times 64$, and the output size after convolution is $64 \times \frac{H}{4} \times \frac{H}{4}$. The processed features are then passed through three branches, each branch contains an Axial Depthwise (AxialDW) convolution, followed by a ResMeVSS block. The ResMeVSS is a residual mamba block (ResMambaBlock) introduced by Ho et al. [25]. Each branch utilizes a different kernel size to capture spatial features at multiple scales, enhancing the generalization ability of the model. The ResMambaBlock further refines global information extraction from the input data.

After passing through the three parallel branches, feature maps from all branches are aggregated and then processed by the Global Average Pooling (GAP). GAP reduces feature dimensionality and extracts global representations. The pooled features are then passed through a linear layer, branching into three separate paths, each corresponding to a distinct attention score. A softmax function is applied to normalize the relationships between three branches. Finally, each branch is weighted by its respective attention score and summed to obtain the final output.

B. Classification-Assisting BAM Block

The BAM Block [26], presented in Fig.3, is utilized to contrast the critical features with the irrelevant ones before evaluating the scores in the classification modules. This block consists of two branches, which are spatial attention and channel attention following the idea of squeezing and exciting. When combined, these two branches could effectively highlight the individual spatial or channel vectors before passing them through the mathematical operations in the next step.

For the channel branch, we apply the GAP (Global Average Pooling) to extract generalized features. The pooled features then pass through two *MLP* layers with ReLU activation function.

$$\mathbf{x_{channel}} = MLP_2(MLP_1(GAP(\mathbf{x_{in}}))) \quad (1)$$

The spatial spatial is equally crucial for learning spatial features. It consists of a series of 2D convolution layers with kernel size 3 × 3 with ReLU and Batch Normalization (BatchNorm) to extract spatial attention across channels.

We define a function f as follow: Denote n be the kernel size of the 2D Convolution, BatchNorm and ReLU are respectively the batch normalization and activation function.

$$f^n(\mathbf{x_{in}}) = BatchNorm(ReLU(Conv2D(\mathbf{x_{in}}))) \quad (2)$$

$$\mathbf{x_{spatial}} = f_3^{1\times1}(f_2^{3\times3}(f_1^{3\times3}(f_0^{1\times1}(\mathbf{x_{in}})))) \quad (3)$$

Finally, the attention block combining both spatial and channel branches to emphasize important features. The final attention output is computed as:

$$\mathbf{x_{attention}} = \mathbf{x_{in}} + \mathbf{x_{in}} \times \mathrm{Sigmoid}(\mathbf{x_{channel}} + \mathbf{x_{spatial}}) \quad (4)$$

C. Spatial Covariance Matrix

After the MAFE and Attention Block, the covariance matrix is utilized to measure the correlation between the query and support sets. The covariance matrix captures second-order statistics which measure the correlation between the query and support sets, unlike the first-order distances such as *cosine* or *euclidean*. Denote $d = H \times W$, $n = K \times d$, and let $\boldsymbol{S_j^i} \in \mathrm{R}^{c\times d}$ be the feature of the j-th support sample of a class, $\mu \in \mathrm{R}^{c\times d}$ be the mean feature presentation of each class. The covariance matrix $\sum \boldsymbol{S^i} \in \mathrm{R}^{c\times c}$ is defined as the following formula:

$$\sum \boldsymbol{S^i} = \frac{1}{n-1}\sum_{j=1}^{K}\left(\boldsymbol{S_j^i} - \mu\right)\left(\boldsymbol{S_j^i} - \mu\right)^T \quad (5)$$

To quantify the correlation between the spatial covariance matrix of each class and the query sample, we use the following function:

$$\boldsymbol{F_i^s} = f\left(\boldsymbol{Q}, \sum \boldsymbol{S^i}\right) = diag\left(\boldsymbol{Q}^T \sum \boldsymbol{S^i}\boldsymbol{Q}\right) \quad (6)$$

A higher value indicates a stronger correlation between the query and support set. Next, we apply a Conv1D layer with a stride of d. This operation ensures that each class receives a final score with the query:

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$$F^s_{\text{score}} = Conv1D(F^s)|stride = d \quad (7)$$

with $F^s = \{F^s_1, F^s_2 \cdots, F^s_C\}$

D. Channel Covariance Matrix

Similar to the Spatial Covariance Matrix, after passing the feature extractor and Attention Block, the channel covariance matrix is utilized to capture the correlation between the channels in the query and support sets. Let $S^j \in R^{c\times d}$ be the feature representation of the j-th support sample in a class, and $\mu \in R^{c\times d}$ be the mean feature of the class. Covariance matrix channel $C^i \in R^{d\times d}$ for class i is expressed as:

$$\sum C^i = \frac{1}{n-1}\sum_{j=1}^{K}(S^i_j - \mu)^T (S^i_j - \mu) \quad (8)$$

The correlation between the channels in the query and support set is defined as:

$$F^C_i = f(Q, \sum C^i) = diag(Q \sum C^i Q^T) \quad (9)$$

Next, a conv1D layer with a stride of c is applied, similar to the case of spatial covariance matrix to obtain the final correlation score between the query sample and support set:

$$F^c_{\text{score}} = Conv1D(F^c)|stride = c \quad (10)$$

where $F^c = \{F^c_1, F^c_2 \cdots, F^c_C\}$

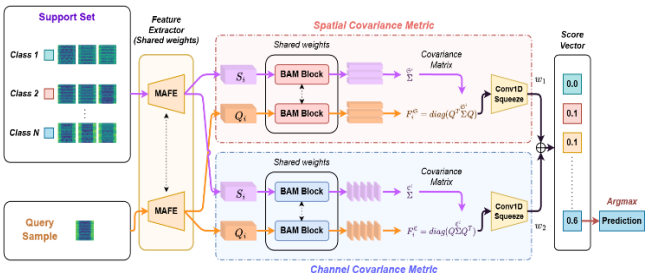

Fig. 1. Images from the Support Set and Query Sample are converted into spectrograms, processed through a shared-weight Feature Extractor, followed by Spatial and Channel Covariance modules, and a Conv1D layer for classification.

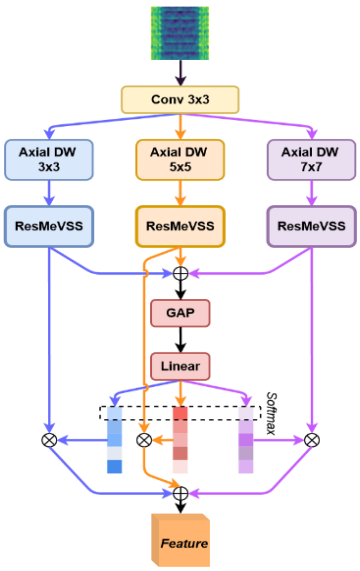

Fig. 2: The proposed MAFE block.

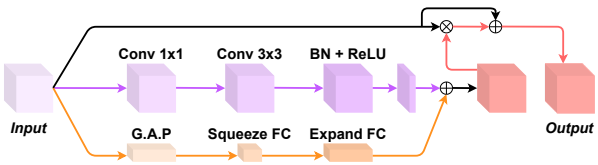

Fig. 3. BAM-based Attention Block.

E. Classification Module

The classification algorithm of a few-shot model serves a crucial role of comparing the query samples with the supporting samples and selecting the best matches. As the model's name "Bi-CovaSC" suggested, there will be two Covariance-matrix-based classification mechanisms focusing on comparing the samples from the spatial-wise and channelwise aspects of the tensors, which leverages the generalization ability of the model's predicting processes. The final score is obtained by weighting the spatial score and the channel score with learnable parameters with the aim of auto-balancing the importance of the two aspects in the training phase:

$$\mathbf{F} = w_1 \times F^s_{\text{score}} + w_2 \times F^c_{\text{score}} \quad (11)$$

Finally, the *Contrastive Loss* is used to push the contrast of the wrong predictions further and pull the correct samples of each class closer to each other. The formula for this loss function is given as below:

$$L = -\ln \frac{\sum e^{F_i|_{i=Y^Q}}}{\sum e^{F_i|_{i=Y^Q}} + \sum e^{F_i|_{i\neq Y^Q}}} \quad (12)$$

IV. Experiment

A. Dataset

To evaluate the performance of the proposed approach for the bearing fault diagnosis task, we use the PU dataset [27]. This dataset was collected under various operating conditions and environments, including both artificial and real-world fault data. Specifically, the PU dataset consists of 13 labels representing different types of faults such as Outer Race, Inner Race, as well as various causes of faults. In this work, as in few-shot learning approaches, for training we create from the PU dataset is divided into two subsets: a support set containing K samples per class, and a query set.

B. Training

To train the proposed model and other compared methods, we use the Adam optimization to optimize the loss in Eq.12. to evaluate and train the model, we use an NVIDIA Tesla T4 GPU with 16GB of memory. The model is initially trained with a learning rate of 0.01, which is reduced by a factor of 10 after every 10 training epochs. To comprehensively assess the proposed method as well as other state-of-the-art approaches in few-shot learning, we conduct experiments with varying numbers of training samples: 195, 260, 650, 1300, and 25,844, similar to [24].

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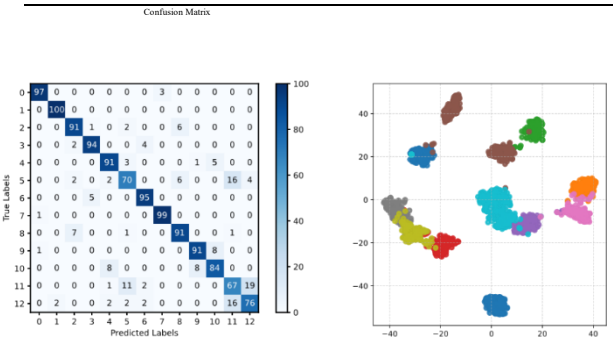
TABLE I. COMPARISON OF ACCURACY, RECALL, AND F1 METRICS (IN %) OF THE PROPOSED MODEL WITH OTHER MODELS WITH VARYING TRAINING SAMPLE IN 1-SHOT CASE.

Method	195 samples			260 samples			650 samples			1300 samples			25844 samples		
	Accuracy	Recall	F1	Accuracy	Recall	F1	Accuracy	Recall	F1	Accuracy	Recall	F1	Accuracy	Recall	F1
RelationNet [14]	67.46	68.05	67.76	71.85	71.78	71.81	80.85	80.72	80.78	83.92	83.93	83.93	88.15	88.85	88.50
SA-CovaMNet [19].	78.54	78.99	78.76	82.38	82.59	82.49	90.38	90.52	90.45	94.31	94.45	94.38	99.23	99.24	99.24
EnsembleNet [17]	80.00	80.21	80.11	88.69	88.72	88.70	95.08	95.11	95.09	97.38	97.38	97.38	99.23	99.23	99.24
MixMamba [24]	83.62	83.79	83.70	88.38	88.38	88.38	95.85	95.93	95.89	97.62	97.64	97.63	99.00	99.01	99.01
Multi-Level [18]	87.38	88.06	87.72	90.23	90.25	90.24	94.31	94.48	94.39	95.85	95.97	95.91	97.31	97.38	97.34
Bi-CovaSC	**89.69**	**89.53**	**89.61**	**93.00**	**93.05**	**93.03**	**96.46**	**96.50**	**96.48**	**97.92**	**97.96**	**97.74**	**99.31**	**99.31**	**99.31**

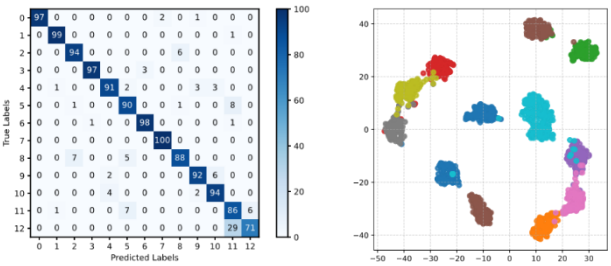
TABLE II. COMPARISON OF ACCURACY, RECALL, AND F1 METRICS (IN %) OF THE PROPOSED MODEL WITH OTHER MODELS WITH VARYING TRAINING SAMPLE IN 5-SHOT CASE.

Method	195 samples			260 samples			650 samples			1300 samples			25844 samples		
	Accuracy	Recall	F1	Accuracy	Recall	F1	Accuracy	Recall	F1	Accuracy	Recall	F1	Accuracy	Recall	F1
RelationNet [14]	77.23	77.48	77.35	82.15	82.18	82.65	86.32	86.34	86.33	91.52	91.38	91.40	95.23	95.24	95.23
SA-CovaMNet [19].	82.15	83.01	82.58	91.31	91.78	91.54	90.15	90.55	90.35	94.00	94.12	94.06	98.85	98.86	98.85
EnsembleNet [17]	87.23	87.05	87.14	89.46	89.74	89.60	96.77	96.81	96.79	98.92	98.93	98.92	99.62	99.62	99.62
MixMamba [24]	90.15	90.28	90.22	92.23	92.23	92.31	97.62	97.62	97.63	99.23	99.23	99.24	99.74	99.74	99.75
Multi-Level [18]	89.46	89.76	89.61	91.23	91.24	91.24	96.00	95.98	95.99	97.00	97.06	97.03	98.85	98.86	98.85
Bi-CovaSC	**92.54**	**92.88**	**92.71**	**95.31**	**95.41**	**95.36**	**98.03**	**98.13**	**98.11**	**99.23**	**99.26**	**99.24**	**99.76**	**99.76**	**99.77**



(a) Confusion Matrix (b) TSNE

Fig. 4: Figures showing the confusion matrix (a) and t-SNE plot (b) for the proposed approach in a 13-class 1-shot scenario, trained with just 195 samples from the PU dataset.



(a) Confusion Matrix (b) TSNE

Fig. 5: Figures showing the confusion matrix (a) and t-SNE plot (b) for the proposed approach in a 13-class 5-shot scenario, trained with just 195 samples from the PU dataset.

The experiments were conducted in two scenarios: a 13-way 1-shot task and a 13-way 5-shot task, with varying amounts of training data, ranging from 195 training samples to 25,844 samples.

C. *Results*

To show the performance of the proposed in the case of as few as 195 training samples, we show the confusion matrix and tsne plots in Fig.4 and Fig.5 for respectively 1-shot and 5-shot scenarios. As shown in the diagonal of the confusion matrices and the class distribution in the tsne plots, the proposed approach can classify effectively the classes of bearing faults even in the case of very limit data.

Furthermore, to evaluate the performance of the proposed model, we conduct the proposed framework and reimplemented some recent few-shot learning based methods proposed for bearing fault diagnosis. The compared methods include: RelationNet [14], SA-CovaMNet [19], EnsembleNet [17], MultiLevel [18], and MixMamba [24]. For a quantitative assessment, we provided the comparison results by varying the training samples with 1-shot and 5-shot scenarios in terms of Accuracy, Recall, and F1 metrics in Tables I and II. From these tables, it is evident that the proposed Bi-CovaSC consistently outperforms other models across all training sample sizes, particularly in larger data regimes (650 samples and above). The model achieves the highest accuracy of 99.74% in the 5-shot scenario with the full dataset (25,844 samples), outperforming other models such as SA-CovaMNet and EnsembleNet.

In less sampling sample scenarios, the proposed Bi-CovaSC shows a noticeable performance advantage over RelationNet and SA-CovaMNet, demonstrating superior generalization ability with limited samples. Specifically, Bi-CovaSC achieves 89.69% Accuracy with 195 samples, while other models remain below 88%. From the tables, Multi-Level and MixMamba models also demonstrate strong performance but falls slightly behind Bi-CovaSC in most cases. Additionally, SA-CovaMNet exhibits stable recall and F1-score improvements across all training sizes, indicating robustness in both minority and majority class detection. These results highlight the effectiveness of Bi-CovaSC in handling few-shot bearing fault diagnosis while maintaining competitive performance across different data scales.

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V. CONCLUSION

In this work, we have introduced a new framework for the task of bearing fault diagnosis in the condition of limited data. The proposed work exploited the newly presented Mamba architecture, leveraging Axial depthwise convolution integration with varying kernel sizes to fuse global and spatial information from the input. In addition, we redesigned the classification module based on the advances of multilevel few-shot approach and the CovamNet to improve the classification performance even in data-scarce scenarios. Our outstanding performance for the PU dataset while compared with state-of-the-arts have demonstrated the effectiveness of the proposed model. In fault bearing diagnosis, data can be affected by noise and environmental factors, leading to variations in fault characteristics for the same type of fault across different machines. Therefore, in the future, we will continue our research to develop a model with high generalization capability, which can be effectively applied to real-world tasks.

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